

PROBLEM STATEMENT

Multi-Layer Nested Chain Rule

Find the derivative with respect to x of the function:

$$y = \sin\{\cos(\log x^3)\}$$

STEP 1 প্রদত্ত ত্রিস্তরীয় সংযোজিত ফাংশনটি উল্লেখ করি (Given Nested Function)

$$y = \sin\{\cos(\log x^3)\}$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [\sin\{\cos(\log x^3)\}]$$

STEP 3 ১ম স্তর: বহিঃস্থ সাইন ফাংশনে চেইন রুল প্রয়োগ করি (Layer 1: Outer Sine Derivative)

$$\frac{dy}{dx} = \cos\{\cos(\log x^3)\} \cdot \frac{d}{dx} [\cos(\log x^3)]$$

STEP 4 ২য় স্তর: মধ্যবর্তী কোসাইন ফাংশনে চেইন রুল প্রয়োগ করি (Layer 2: Cosine Derivative)

$$\frac{d}{dx} [\cos(\log x^3)] = -\sin(\log x^3) \cdot \frac{d}{dx} (\log x^3)$$

STEP 5 ৩য় স্তর: অভ্যন্তরীণ লগারিদম ও ঘাত অংশের অন্তরক নির্ণয় করি (Layer 3: Logarithm Derivative)

$$\frac{d}{dx} (\log x^3) = \frac{1}{x^3} \cdot \frac{d}{dx} (x^3) = \frac{1}{x^3} \cdot 3x^2 = \frac{3}{x}$$

STEP 6 প্রতিটি স্তরের অন্তরক একত্রিত করে চূড়ান্ত সরল রূপ পাই (Combining All Layers)

$$\frac{dy}{dx} = \cos\{\cos(\log x^3)\} \cdot [-\sin(\log x^3)] \cdot \left(\frac{3}{x}\right) = -\frac{3}{x} \sin(\log x^3) \cos\{\cos(\log x^3)\}$$

FINAL ANSWER
$$\frac{dy}{dx} = -\frac{3}{x} \sin(\log x^3) \cos\{\cos(\log x^3)\}$$

🔴 BOX 1: FORMULATION & MULTI-LAYER CHAIN RULE

- **Triple Composition Rule:** If $y = f(g(h(x)))$, then:

$$\frac{dy}{dx} = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$$

- Here, $f(u) = \sin u$, $g(v) = \cos v$, $h(x) = \log(x^3) = 3 \log x$.
- $f'(u) = \cos u$, $g'(v) = -\sin v$, $h'(x) = \frac{3}{x}$.
- Multiplying sequentially yields: $-\frac{3}{x} \sin(\log x^3) \cos(\cos(\log x^3))$.

⚠️ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Missing Negative Sign from Cosine:** The derivative of $\cos(\dots)$ is $-\sin(\dots)$. Missing this minus sign loses substantial accuracy marks.
- **Incomplete Chain on $\log x^3$:** Writing $\frac{d}{dx}(\log x^3) = \frac{1}{x^3}$ instead of $\frac{3x^2}{x^3} = \frac{3}{x}$.
- **Corrupting Nested Brackets:** Do not mix up the arguments: $\cos(\cos(\dots))$ must be preserved intact.

🌐 BOX 3: REAL DOMAIN & LOGARITHMIC RESTRICTIONS

- **Domain of Definition:** The argument of $\log(x^3)$ requires $x^3 > 0 \iff x > 0$. Thus:

$$\text{Domain} = (0, \infty)$$

- **Differentiability:** Continuous and infinitely differentiable on $(0, \infty)$.
- **Singular Behavior at Origin:** As $x \rightarrow 0^+$, the factor $\frac{3}{x} \rightarrow +\infty$, causing rapid unbounded oscillations.

💡 BOX 4: LEIBNIZ STEPWISE SUBSTITUTION

- Let $w = \log x^3 = 3 \log x \implies \frac{dw}{dx} = \frac{3}{x}$.
- Let $v = \cos w \implies \frac{dv}{dw} = -\sin w = -\sin(\log x^3)$.
- Let $y = \sin v \implies \frac{dy}{dv} = \cos v = \cos(\cos(\log x^3))$.
- By Leibniz Chain Formula:

$$\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{dw} \cdot \frac{dw}{dx} = -\frac{3}{x} \sin(\log x^3) \cos(\cos(\log x^3))$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 4 / 5 MARKS)

- **1.0 Mark:** Correct differentiation of outer sine term $\cos(\cos(\log x^3))$.
- **1.0 Mark:** Correct differentiation of middle cosine term $-\sin(\log x^3)$.
- **1.0 Mark:** Accurate chain rule computation of $\frac{d}{dx}(\log x^3) = \frac{3}{x}$.
- **1.0 Mark:** Proper negative sign handling and final boxed mathematical assembly.